

Span composition using fake pullbacks

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Abstract

The construction of a category of spans can be made in some categories $\mathcal{C}$ which do not have pullbacks in the traditional sense. The PROP for monoids is a good example of such a $\mathcal{C}$. The 2012 book on Homological Algebra by Marco Grandis gives the proof of associativity of relations in a Puppe-exact category based on a 1967 paper of M.Š. Calenko. The proof here is a restructuring of that proof in the spirit of the first sentence of this Abstract.

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Introduction

The construction of a category of spans can be made in some categories $\mathcal{C}$ not having pullbacks in the traditional sense, only having some form of *fake pullback*. The PROP for monoids is a good example of such a $\mathcal{C}$; it has a forgetful functor to the category of finite sets which takes fake pullbacks to genuine pullbacks.

As discussed in the book [7] by Marco Grandis, relations in a Puppe-exact category $\mathcal{C}$ are zig-zag diagrams of monomorphisms and epimorphisms, not just jointly monomorphic spans as for a regular category (see [4] for example). Associativity of these zig-zag relations was proved by M.Š. Calenko [8] over 50 years ago.

The proof in the present paper is a restructuring of that proof in the spirit of fake pullbacks. The original category $\mathcal{C}$ does not even need to be pointed, but it should have a *suitable factorization system* $(\mathcal{E}, \mathcal{M})$. The fake pullbacks are constructed in what we call $\text{Spn}(\mathcal{E}, \mathcal{M})$, not in $\mathcal{C}$ itself, and there is no forgetful functor turning them into genuine pullbacks. The relations are spans in $\text{Spn}(\mathcal{E}, \mathcal{M})$. The main point in proving associativity of the span composition is that fake pullbacks stack properly.

1 Suitable factorization systems

Let $(\mathcal{E}, \mathcal{M})$ be a factorization system in the sense of [5] on a category $\mathcal{C}$. That is, $\mathcal{E}$ and $\mathcal{M}$ are sets of morphisms of $\mathcal{C}$ which contain the isomorphisms, are closed under composition, and satisfy the conditions:

FS1. if $mu = ve$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$ then there exists a unique w with $we = u$ and $mw = v$;

FS2. every morphism f factorizes $f = me$ with $e \in \mathcal{E}$ and $m \in \mathcal{M}$.

If we write $f : A \twoheadrightarrow B$, we mean $f \in \mathcal{E}$. If we write $f : A \rightharpoonup B$, we mean $f \in \mathcal{M}$.

The factorization system $(\mathcal{E}, \mathcal{M})$ is *suitable* when it satisfies:

SFS1. pullbacks of arbitrary morphisms along members of $\mathcal{M}$ exist;

SFS2. pushouts of arbitrary morphisms along members of $\mathcal{E}$ exist;

SFS3. the pullback of an $\mathcal{E}$ along an $\mathcal{M}$ is in $\mathcal{E}$;

SFS4. the pushout of an $\mathcal{M}$ along an $\mathcal{E}$ is in $\mathcal{M}$;

SFS5. a commutative square of the form

$$\begin{array}{ccc} D & \xrightarrow{n} & B \\ e \downarrow & & \downarrow d \\ A & \xrightarrow{m} & C \end{array}$$

is a pullback if and only if it is a pushout.

Proposition 1. (i) *Spans of the form $X \leftarrow S \rightarrow Y$ are jointly monomorphic.*

(ii) *Cospans of the form $X \rightarrow C \leftarrow Y$ are jointly epimorphic.*

Proof. A pushout of $X \leftarrow S \rightarrow Y$ exists by SFS2 and $X \leftarrow S \rightarrow Y$ is the pullback of the resultant cospan by SFS5. Pullback spans are jointly monomorphic. This proves (i), and (ii) is dual. $\square$

Example 1. 1. Take $\mathcal{C}$ to be the category of groups $\mathbf{Grp}$, $\mathcal{E}$ to be the set of surjective morphisms and $\mathcal{M}$ to be the set of injective morphisms.

2. Take $\mathcal{C}$ to be any Puppe-exact category as studied by Grandis [7], $\mathcal{E}$ the epimorphisms and $\mathcal{M}$ the monomorphisms. This includes all abelian categories.

3. Take $\mathcal{C}$ to be the category $\mathbf{Spn}(\mathbf{Set}_{\text{inj}})$ of spans in the category of sets and injective functions, $\mathcal{E}$ the i^* and $\mathcal{M}$ the i_* .

4. Take $\mathcal{C}$ to be any groupoid with $\mathcal{E} = \mathcal{M}$ containing all morphisms.

Now we remind the reader of Lemma 2.5.9 from [7].

Lemma 2. *In a commutative diagram of the form*

$$\begin{array}{ccccc}
 A & \xrightarrow{d} & B & \xrightarrow{i} & C \\
 \ell \downarrow & & \downarrow m & & \downarrow n \\
 X & \xrightarrow{e} & Y & \xrightarrow{j} & Z
 \end{array}, \tag{1.1}$$

the horizontally pasted square is a pullback if and only if both the component squares are pullbacks.

Proof. “If” is true without any condition on the morphisms. For the converse, using SFS1, take the pullback of j and n to obtain another pastable pair of squares with the same left, right and bottom sides. The top composites are equal. By factorization system properties and SFS3, the new top is also a factorization of $i \circ d$ and thus isomorphic to the given factorization. So both of the old squares are also pullbacks. $\square$

We might call the diagram (1.1) an $\mathcal{M}$ -morphism of factorizations. The dual of the lemma concerns pushouts in $\mathcal{E}$ -morphisms of factorizations. We did not use SFS5 in proving the Lemma, however that condition does tell us that the left square of (1.1) is also a pushout when the pasted diagram is a pullback.

2 The bicategory of EM-spans

Let $(\mathcal{E}, \mathcal{M})$ be a suitable factorization system on the category $\mathcal{C}$.

We define a bicategory $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$ with the same objects as $\mathcal{C}$. The morphisms $(d, R, m) : U \rightarrow W$ are spans $U \leftarrow R \rightarrow W$ in $\mathcal{C}$. The 2-cells are the usual morphisms of spans. Composition is the usual composition of spans; this uses conditions SFS1, SFS3 and closure of $\mathcal{E}$ under composition.

Each $(X \xrightarrow{m} Y) \in \mathcal{M}$ gives a morphism $m_* : X \xrightarrow{(1_X, X, m)} Y$ in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$ and each $(X \xrightarrow{e} Y) \in \mathcal{E}$ gives a morphism $e^* : Y \xrightarrow{(e, X, 1_X)} X$ in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$. Write $\mathcal{M}_*$ for the class of all m_* with $m \in \mathcal{M}$ and write $\mathcal{E}^*$ for the class of all e^* with $e \in \mathcal{E}$.

Every morphism $(d, R, m) : U \rightarrow W$ decomposes as $U \xrightarrow{d^*} R \xrightarrow{m_*} W$; this decomposition $\mathcal{M}_* \mathcal{E}^*$ is unique up to isomorphism. Using condition SFS5, we see that every morphism $(d, R, m) : U \rightarrow W$ also decomposes as $U \xrightarrow{m'_*} Q \xrightarrow{e'^*} W$ by taking the pushout $U \xrightarrow{m'} Q \xleftarrow{d'} W$ of $U \xleftarrow{d} R \xrightarrow{m} W$; this decomposition $\mathcal{E}^* \mathcal{M}_*$ is also unique up to isomorphism.

Proposition 1 tells us that the bicategory $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$ is locally preordered.

Proposition 3. *Given $X \xrightarrow{m_*} Y \in \mathcal{M}_*$ and $Z \xrightarrow{e^*} Y \in \mathcal{E}^*$, there exists a diagram of the form*

$$\begin{array}{ccc} J & \xrightarrow{\bar{e}^*} & X \\ \bar{m}_* \downarrow & \Leftarrow & \downarrow m_* \\ Z & \xrightarrow{e^*} & Y \end{array} \quad (2.2)$$

in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$, with $\bar{e} \in \mathcal{E}$ and $\bar{m} \in \mathcal{M}$, which is unique up to isomorphism.

Proof. Interpreting $m_* \circ \bar{e}^* \leq e^* \circ \bar{m}_*$, we see that $\bar{m} \circ \bar{e}$ is forced to be an $(\mathcal{E}, \mathcal{M})$ factorization of $e \circ m$. $\square$

Proposition 4. *If $m \in \mathcal{M}$ then m_* is a discrete fibration in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$; that is, each functor $\mathrm{Spn}(\mathcal{E}, \mathcal{M})(K, m_*)$ is a discrete fibration. If $e \in \mathcal{E}$ then e^* is a codiscrete cofibration in $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$; that is, each functor $\mathrm{Spn}(\mathcal{E}, \mathcal{M})(e^*, K)$ is a discrete fibration.*

Write $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$ for the classifying category of the bicategory $\mathrm{Spn}(\mathcal{E}, \mathcal{M})$; that is, it has the same objects while the morphisms $[d, R, m] : U \rightarrow W$ are the isomorphism classes of morphism $(d, R, m) : U \rightarrow W$.

Proposition 5. *Both $(\mathcal{E}^*, \mathcal{M}_*)$ and $(\mathcal{M}_*, \mathcal{E}^*)$ are factorization systems on the category $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$.*

Proposition 6. *Pullbacks in $\mathcal{C}$ whose morphisms are all in $\mathcal{M}$ are taken by $(-)_*$ to pullbacks in $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$. Also, pushouts in $\mathcal{C}$ whose morphisms are all in $\mathcal{E}$ are taken by $(-)^*$ to pullbacks in $\mathrm{Spn}[\mathcal{E}, \mathcal{M}]$.*

3 Relations as spans of spans

By regular categories we mean those in the sense of Barr [1] which admit all finite limits. One characterization of the bicategory of relations in a regular category was given in [4]. A relation from X to Z in a regular category is a jointly monomorphic span from X to Z ; these are composed using span composition followed by factorization. Equivalently, a relation from X to Z is a subobject of $X \times Y$.

The category $\mathbf{Grp}$ of groups is regular. So relations are subgroups of products $X \times Z$. The Goursat Lemma [6] is a bijection between subgroups $S \leq X \times Z$ of a cartesian product of groups X and Z and end-fixed isomorphism classes of diagrams

$$X \xleftarrow{m} U \xrightarrow{d} Y \xleftarrow{e} V \xrightarrow{n} Z . \quad (3.3)$$

To obtain S from (3.3), take the pullback $U \xleftarrow{\bar{e}} P \xrightarrow{\bar{d}} V$ of $U \xrightarrow{d} Y \xleftarrow{e} V$ then S is the image of $P \xrightarrow{(m\bar{e}, e\bar{d})} X \times Y$. To obtain the zig-zag (3.3) from $S \hookrightarrow X \times Z$, factorize the two restricted projections to obtain

$$X \xleftarrow{m} U \xleftarrow{e'} S \xrightarrow{d'} V \xrightarrow{n} Z ,$$

then pushout e' and d' to obtain d and e .

This motivates the definition of *relation* from X to Z in a category $\mathcal{C}$ equipped with a suitable factorization system $(\mathcal{E}, \mathcal{M})$ as an isomorphism class of diagrams of the form (3.3). A good reference is [7] for the case where $\mathcal{C}$ is Puppe-exact.

The starting point for the present paper was the simple observation that a relation diagram (3.3) is a span in $\mathbf{Spn}(\mathcal{E}, \mathcal{M})$:

$$(d, U, m, Y, e, V, n) : X \xleftarrow{(d, U, m)} Y \xrightarrow{(e, V, n)} Z . \quad (3.4)$$

We would like to define $\mathbf{Rel}[\mathcal{E}, \mathcal{M}]$ to be the classifying category of the bicategory $\mathbf{Spn}(\mathbf{Spn}[\mathcal{E}, \mathcal{M}])$. This is satisfactory as a definition of the graph but for the composition we need a way to compose spans in $\mathbf{Spn}[\mathcal{E}, \mathcal{M}]$.

4 A fake pullback construction

Let $(\mathcal{E}, \mathcal{M})$ be a suitable factorization system on a category $\mathcal{C}$. Although $\mathcal{C}$ may not have all pullbacks, we will now show that $\mathbf{Spn}[\mathcal{E}, \mathcal{M}]$ does have some kind of span composition and this gives a composition of relations. The construction and proof of associativity restructures that of [8]. However, we will see in Section 5 that the properties of $\mathbf{Spn}(\mathcal{E}, \mathcal{M})$ established in Section 2 allow an abstract proof of associativity of composition of relations.

Take any cospan $U \xrightarrow{(d,R,m)} W \xleftarrow{(e,S,n)} V$ in $\text{Spn}(\mathcal{E}, \mathcal{M})$. Construct the diagram

$$\begin{array}{ccccc}
 Q & \xleftarrow{s} & Y & \xrightarrow{j} & V \\
 \uparrow r & & \uparrow \bar{e} & & \uparrow e \\
 X & \xleftarrow{\bar{d}} & Z & \xrightarrow{\bar{m}} & S \\
 \downarrow i & & \downarrow \bar{n} & & \downarrow n \\
 U & \xleftarrow{d} & R & \xrightarrow{m} & W
 \end{array} \tag{4.5}$$

in which the bottom right square is a pullback of $R \rightrightarrows W \leftarrow S$, the bottom left square is an $(\mathcal{E}, \mathcal{M})$ -factorization of the composite $Z \rightrightarrows R \rightrightarrows U$, the top right square is an $(\mathcal{E}, \mathcal{M})$ -factorization of the composite $Z \rightrightarrows S \rightrightarrows V$, and the top left square is a pushout of the span $X \leftarrow Z \rightrightarrows Y$.

We call the span $U \xleftarrow{(r,R,i)} Q \xrightarrow{(s,S,j)} V$ the *fake pullback* of the given cospan $U \xrightarrow{(d,R,m)} W \xleftarrow{(e,S,n)} V$. We obtain the diagram (4.6) in $\text{Spn}(\mathcal{E}, \mathcal{M})$. The top left square comes from a pushout in $\mathcal{C}$, the bottom right square from a pullback in $\mathcal{C}$, while the 2-cells come from factorizing an $\mathcal{M}$ followed by an $\mathcal{E}$ as an $\mathcal{E}$ followed by an $\mathcal{M}$.

$$\begin{array}{ccccc}
 Q & \xrightarrow{s^*} & Y & \xrightarrow{j^*} & V \\
 \downarrow r^* & \lrcorner & \downarrow \bar{e}^* & \xRightarrow{\sigma} & \downarrow e^* \\
 X & \xrightarrow{\bar{d}^*} & Z & \xrightarrow{\bar{m}^*} & S \\
 \downarrow i^* & \xleftarrow{\rho} & \downarrow \bar{n}^* & \lrcorner & \downarrow n^* \\
 U & \xrightarrow{d^*} & R & \xrightarrow{m^*} & W
 \end{array} \tag{4.6}$$

Note that, if d and m are invertible, then so are s and j .

Next we will provide some switching rules which allow us to see that fake pullbacks stack.

In diagram (4.7), we have taken composable pullbacks in $\mathcal{C}$ with morphisms as marked on the left-hand side, then factored the top and bottom composites to obtain the right-hand side.

$$\begin{array}{ccc}
 \begin{array}{ccccc}
 A & \xrightarrow{i} & B & \xrightarrow{d} & C \\
 \downarrow \ell & \lrcorner & \downarrow m & \lrcorner & \downarrow n \\
 X & \xrightarrow{j} & Y & \xrightarrow{e} & Z
 \end{array} & = & \begin{array}{ccccc}
 A & \xrightarrow{\bar{d}} & I & \xrightarrow{\bar{i}} & C \\
 \downarrow \ell & & \downarrow h & & \downarrow n \\
 X & \xrightarrow{\bar{e}} & J & \xrightarrow{\bar{j}} & Z
 \end{array}
 \end{array} \tag{4.7}$$

By Lemma 2, both squares on the right-hand side are pullbacks; indeed, the left-hand square on the right-hand side of the equation is also a pushout (by SFS5).

We can write this in $\text{Spn}(\mathcal{E}, \mathcal{M})$ as a *pullback/2-cell interchange equation* (4.8).

$$\begin{array}{ccc}
 I & \xrightarrow{h_*} & J & \xrightarrow{\bar{e}^*} & X \\
 \bar{i}_* \downarrow & \lrcorner & \bar{j}_* \downarrow & \Leftarrow & \downarrow j_* \\
 C & \xrightarrow{n_*} & Z & \xrightarrow{e^*} & Y
 \end{array}
 =
 \begin{array}{ccc}
 I & \xrightarrow{\bar{d}^*} & A & \xrightarrow{\ell_*} & X \\
 \bar{i}_* \downarrow & \Leftarrow & \downarrow i_* & \lrcorner & \downarrow j_* \\
 C & \xrightarrow{d^*} & B & \xrightarrow{m_*} & Y
 \end{array}
 \quad (4.8)$$

Dually, in diagram (4.9), we have taken composable pushouts in $\mathcal{C}$ with morphisms as marked on the left-hand side, then factored the top and bottom composites to obtain the right-hand side.

$$\begin{array}{ccc}
 A & \xrightarrow{i} & B & \xrightarrow{p} & C \\
 d \downarrow & \lrcorner & e \downarrow & \lrcorner & \downarrow f \\
 X & \xrightarrow{j} & Y & \xrightarrow{q} & Z
 \end{array}
 =
 \begin{array}{ccc}
 A & \xrightarrow{\bar{p}} & I & \xrightarrow{\bar{i}} & C \\
 d \downarrow & & \downarrow g & & \downarrow f \\
 X & \xrightarrow{\bar{q}} & J & \xrightarrow{\bar{j}} & Z
 \end{array}
 \quad (4.9)$$

By the dual of Lemma 2, both squares on the right-hand side are pushouts; indeed, the right-hand square on the left-hand side of the equation is also a pullback (by SFS5). In $\text{Spn}(\mathcal{E}, \mathcal{M})$ this gives a *pushout/2-cell interchange equation* (4.10).

$$\begin{array}{ccc}
 J & \xrightarrow{\bar{j}^*} & Z & \xrightarrow{f^*} & C \\
 \bar{q}^* \downarrow & \Rightarrow q^* \downarrow & \lrcorner & \downarrow p^* \\
 X & \xrightarrow{j_*} & Y & \xrightarrow{e^*} & B
 \end{array}
 =
 \begin{array}{ccc}
 J & \xrightarrow{g^*} & I & \xrightarrow{\bar{i}^*} & C \\
 \bar{q}^* \downarrow & \lrcorner & \downarrow \bar{p}^* \Rightarrow & \downarrow p^* \\
 X & \xrightarrow{d^*} & A & \xrightarrow{i_*} & B
 \end{array}
 \quad (4.10)$$

Now, in the diagram (4.11) of two horizontally stacked fake pullbacks, we can apply a pushout/2-cell interchange in the middle two top squares and a pullback/2-cell interchange in the middle bottom two squares.

$$\begin{array}{ccccccc}
 Q & \xrightarrow{s^*} & Y & \xrightarrow{j_*} & V & \xrightarrow{s'^*} & A & \xrightarrow{j'_*} & D \\
 r_* \downarrow & \lrcorner & \bar{e}^* \downarrow & \Rightarrow e^* \downarrow & \lrcorner & \bar{e}'^* \downarrow & \Rightarrow & \downarrow e'^* \\
 X & \xrightarrow{\bar{d}^*} & Z & \xrightarrow{\bar{m}_*} & S & \xrightarrow{\bar{d}'^*} & B & \xrightarrow{\bar{m}'_*} & E \\
 i_* \downarrow & \Leftarrow & \bar{n}_* \lrcorner & \downarrow n_* \Leftarrow & \bar{n}'_* \lrcorner & \downarrow n'_* \\
 U & \xrightarrow{d^*} & R & \xrightarrow{m_*} & W & \xrightarrow{d'^*} & C & \xrightarrow{m'_*} & F
 \end{array}
 \quad (4.11)$$

The result is another fake pullback. To see this, we take the pullback $R \xleftarrow{\bar{p}} P' \xrightarrow{\ell'} C$ of $R \xrightarrow{m} W \xleftarrow{d'} C$ and the pullback $P' \xleftarrow{\bar{n}} P \xrightarrow{\ell} B$ of $P' \xrightarrow{\ell'} C \xleftarrow{\bar{n}'} B$ to obtain both an equation

$$\begin{array}{ccc}
 P & \xrightarrow{\bar{n}} & P' & \xrightarrow{\bar{p}} & R \\
 \ell \downarrow & \lrcorner & \ell' \downarrow & \lrcorner & \downarrow m \\
 B & \xrightarrow{\bar{n}'} & C & \xrightarrow{d'} & W
 \end{array}
 =
 \begin{array}{ccc}
 P & \xrightarrow{p} & Z & \xrightarrow{\bar{n}} & R \\
 \ell \downarrow & & \downarrow \bar{m} & & \downarrow m \\
 B & \xrightarrow{\bar{d}'} & S & \xrightarrow{n} & W
 \end{array}$$

of the same form as (4.7) and, using condition (SFS), an equation

$$\begin{array}{ccc}
 P \rightharpoonup \xrightarrow{\ell} B & \xrightarrow{\bar{e}'} & A \\
 p \downarrow \lrcorner & \lrcorner & \downarrow s' \\
 Z \rightharpoonup \xrightarrow{\bar{m}} S & \xrightarrow{e} & V
 \end{array}
 =
 \begin{array}{ccccc}
 P & \xrightarrow{f} & I & \xrightarrow{k} & A \\
 p \downarrow \lrcorner & & \downarrow q & & \downarrow s' \\
 Z & \xrightarrow{\bar{e}} & Y & \xrightarrow{j} & V
 \end{array}$$

of the same form as (4.9). So (4.11) can be rewritten as a fake pullback as follows.

$$\begin{array}{ccccccccc}
 Q & \xrightarrow{s^*} & Y & \xrightarrow{q^*} & I & \xrightarrow{k_*} & A & \xrightarrow{j'_*} & D \\
 r_* \downarrow \lrcorner & \lrcorner & \bar{e}^* \downarrow \lrcorner & \lrcorner & f^* \Rightarrow & \lrcorner & \bar{e}'^* \Rightarrow & \lrcorner & e'^* \downarrow \\
 X & \xrightarrow{\bar{d}^*} & Z & \xrightarrow{p^*} & P & \xrightarrow{\ell_*} & B & \xrightarrow{\bar{m}'_*} & E \\
 i_* \downarrow \lrcorner & \lrcorner & \bar{n}_* \Leftarrow & \lrcorner & \bar{n}_* \lrcorner & \lrcorner & \bar{n}'_* \lrcorner & \lrcorner & n'_* \downarrow \\
 U & \xrightarrow{d^*} & R & \xrightarrow{\bar{p}^*} & P' & \xrightarrow{\ell'_*} & C & \xrightarrow{m'_*} & F
 \end{array}$$

Composition in $\text{Rel}[\mathcal{C}, \mathcal{M}]$ is now obtained as for spans in any category with pullbacks except that we use the fake pullbacks. The zig-zag diagram in $\mathcal{C}$ with all morphisms identities of X represents the identity relation of X . The stacking property shows that composition is associative. We conclude that $\text{Rel}[\mathcal{C}, \mathcal{M}]$ is a category.

5 An abstraction

Let $\mathcal{S}$ be a locally preordered bicategory.

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